

Milestone 2: KPI Engineering and Dashboard Planning (Weeks 3–4)

1. Introduction

Milestone 2 marks the transition from data preparation to analytical enhancement and dashboard conceptualization. After successfully scraping, cleaning, and structuring military data in Milestone 1, the focus of this stage is to convert raw data into meaningful analytical indicators and prepare the logical foundation for interactive dashboards.

This milestone ensures that the dataset is not only clean but also analytically valuable. Raw numbers alone do not support strategic insights; therefore, derived metrics and contextual enrichment are necessary.

Milestone 2 consists of two modules:

- Module 3: KPI Feature Engineering
- Module 4: Dashboard Planning and Prototyping

These modules bridge the gap between data engineering and visualization.

2. Module 3: KPI Feature Engineering

2.1 Objective

The primary objective of KPI Feature Engineering is to transform cleaned military data into meaningful indicators that support comparative and strategic analysis. While base metrics such as number of tanks or aircraft provide descriptive information, KPIs enable normalized comparisons across countries of varying sizes and economic capacities.

This module focuses on:

- Creating derived indicators
- Normalizing military metrics
- Adding contextual metadata
- Preparing BI-ready datasets

2.2 Rationale for KPI Development

Military capability cannot be evaluated using raw counts alone. Larger nations naturally possess more assets and personnel, which can distort comparisons. Therefore, KPIs were engineered to normalize and contextualize values.

Key analytical goals included:

- Fair comparison between nations
- Efficiency measurement
- Strategic positioning
- Coalition contribution estimation

2.3 KPI Computation Process

KPIs were computed using Python to ensure reproducibility and automation. Each KPI was derived using mathematically valid formulas and validated for accuracy.

2.3.1 Assets per Capita

Formula:

Total Military Assets / Total Population

Purpose:

Measures militarization relative to population size.

Analytical Value:

Allows comparison between highly populated and smaller nations.

2.3.2 Budget-to-GDP Ratio

Formula:

Defense Budget / Purchasing Power Parity

Purpose:

Indicates economic prioritization of defense.

Analytical Value:

Highlights strategic investment levels.

2.3.3 Power Index Rank Gap

Formula:

Country Rank – Top Rank

Purpose:

Measures distance from global leader.

Analytical Value:

Helps evaluate relative military standing.

2.3.4 Budget per Active Personnel

Formula:

Defense Budget / Active Personnel

Purpose:

Measures financial investment per soldier.

Analytical Value:

Distinguishes modernization-focused forces from manpower-heavy ones.

2.3.5 Air–Land–Naval Balance Ratio

Derived from:

- Aircraft counts
- Tank counts
- Naval fleet size

Purpose:

Analyzes domain balance.

Analytical Value:

Identifies doctrinal focus.

2.4 Metadata Enrichment

To support geopolitical grouping, additional contextual fields were added.

2.4.1 Region Mapping

Countries were mapped to geographic regions using a standardized dictionary in Python.

Purpose: Supports regional comparisons and coalition analysis.

2.4.2 Continent Classification

Continental labels were added.

Purpose: Enables macro-level grouping.

2.4.3 NATO Membership Flag

Binary indicator (1/0).

Purpose: Allows alliance-based analysis.

2.5 Data Structuring**The dataset was prepared for BI compatibility:**

- Numeric types standardized
- Column names normalized
- Wide format retained for aggregation

The final dataset was exported as: military_final.xlsx

All transformations were scripted in: generate_kpis.py

3. Module 4: Dashboard Planning and Prototyping**3.1 Objective**

This module focuses on conceptual design to ensure dashboards are analytically meaningful and user-friendly.

3.2 Dashboard Architecture**Four dashboards were conceptualized:**

- Quick Stats
- Nation Overview
- Compare Powers
- Coalition Builder

Each dashboard serves a distinct analytical purpose.

3.3 Interaction Design

- Country filters
- Region filters
- Alliance filters
- Navigation buttons
- Cross-filtering

3.4 Prototype Validation

Sample visuals were built to test:

- KPI responsiveness
- Aggregation logic
- Filter behavior

This reduced development risks.

4. Milestone 2 Summary

Milestone 2 successfully converts cleaned data into analytical intelligence. KPIs and metadata enrich the dataset, while dashboard planning ensures clarity and usability.

Outputs include:

- KPI-enhanced dataset
- Metadata enrichment
- Dashboard blueprint
- Validated interaction logic

This milestone prepares the project for full dashboard implementation.